KARISHNII KUBER

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SUMMARY

Computer Science undergraduate with a broad technical range spanning AI/ML, backend engineering, distributed systems, and GPU-accelerated computing. Fine-tuned Transformer models and built RAG pipelines with PyTorch and HuggingFace. Currently building production microservices and Salesforce CRM integrations at Aptos Retail. Achieved 17,000× speedup on a GPU ray tracer using CUDA. Two international conference publications. Strong in Python, C++, Java, Go, and TypeScript.

EDUCATION

PES University Expected May 2026

B.Tech, Computer Science and Engineering | CGPA: 8.66/10 | CNR Rao Merit Scholarship (Top 20%) *Bengaluru, India*

Coursework: Machine Learning, Large Language Models, DSA, OS, Computer Networks, Heterogeneous Parallelism, Big Data

PUBLICATIONS

- LoRaWAN Train Tracking & Collision Detection — ICICT London, 2026 (Intl. Conf. on Information & Communication Technology)
- AI-Powered Placement Portal — ICTIS, 2026 (Intl. Conf. on Technology & Information Systems)

EXPERIENCE

Software Engineering Intern — Aptos Retail Jan 2026 – Present

Extensibility Team, Bengaluru

- Developing backend microservices, REST APIs, and integration tooling using Hapi.js, TypeScript, and Docker for the Aptos One retail platform extensibility layer.
- Building Salesforce CRM integration for Aptos One, working with CRM architecture, API integration patterns, and cross-system data flows.
- Building event-driven architectures with RabbitMQ and deploying scalable services on AWS for enterprise retail clients.
- Collaborating in Agile sprints with cross-functional teams; participating in code reviews and following secure coding practices.

Teaching Assistant, Python Programming — PES University Aug – Dec 2025

- Mentored 60+ undergraduates through weekly review sessions focused on debugging, algorithmic problem-solving, and Python best practices.
- Assisted faculty in designing lab exercises and evaluating student submissions.

Website Designer — Infobells Jun – Aug 2024

- Designed and deployed a responsive e-commerce website (WordPress + Hostinger), improving public visibility and driving daily engagement with 100s of buyers.
- Managed end-to-end delivery including domain setup, hosting configuration, SEO optimization, and post-launch analytics.

PROJECTS

Sentinel AI — LLM Security Guardrail Suite

PyTorch, HuggingFace Transformers, DeBERTa-v3, BERT NER, FAISS, BM25, Flan-T5, Streamlit | GitHub

- Fine-tuned DeBERTa-v3-small for 4-class adversarial prompt classification (safe/jailbreak/injection/toxic) using Focal Loss, temperature calibration, and gradient accumulation, achieving 0.81 macro F1 on 2,646 merged samples from 4 HuggingFace datasets.
- Built hybrid RAG pipeline with FAISS dense retrieval (MiniLM-L6-v2) + BM25 sparse retrieval, Reciprocal Rank Fusion, cross-encoder reranking, and Flan-T5-base chain-of-thought generation. Achieved 95% retrieval recall across 11 policy documents (OWASP, GDPR, NIST AI RMF, MITRE ATLAS, etc.).
- Developed PII detector combining Transformer NER (BERT-base-NER) with 12 regex patterns, Luhn validation, and IP octet checks.
- Ran adversarial robustness evaluations across 6 perturbation types (typo injection, leetspeak, unicode homoglyphs, whitespace insertion, prefix noise).

LoRaWAN Train Tracking & Collision Detection

Python, Apache Kafka, Apache Flink, Redis, PyTorch (LSTM), X25519, AES-GCM | GitHub

- Architected a real-time distributed telemetry pipeline processing continuous event streams with sub-800ms end-to-end latency.
- Designed a scalable event-driven backend using Kafka and Flink for stream analytics, state management, and fault tolerance.
- Developed a PyTorch LSTM model for real-time collision detection with sub-second inference latency.
- Implemented secure communication using X25519 key exchange and AES-GCM encryption for telemetry integrity.

RayTraceX — GPU-Accelerated Ray Tracer

C++, CUDA, OpenMP, BVH, Metal/Swift | GitHub

- Achieved 17,000× speedup (13 min → 45ms per frame) through BVH spatial indexing and hybrid CPU-GPU architecture.

- Developed and benchmarked 4 implementations (CPU, BVH-optimized, Metal GPU, heterogeneous CPU-GPU), demonstrating systematic performance tuning and memory access optimization across platforms.

Invoice Management System — Production Web Application
React 18, TypeScript, Vite, Tailwind CSS, Node.js, Express, PostgreSQL, Prisma, JWT, Docker | Live
- Automated the complete invoice tracking and management workflow for a small business, from daily field orders to company-wide invoice oversight with field report tracking.
- Implemented RBAC (3 user roles), batch operations, backdate approval workflows, audit trails, and one-click CSV export with server-side filtering.
- Deployed on Vercel (frontend) and Railway (backend with Docker) with Prisma ORM and JWT-based authentication. Used daily in production.

AI-Powered Placement Portal
Go, React/TypeScript, PostgreSQL, Mistral LLM, Ollama | GitHub
- Built scalable Go backend with RESTful microservices; orchestrated concurrent API integrations (GitHub, LeetCode) using goroutines for non-blocking data aggregation.
- Deployed Mistral LLM via Ollama for AI-driven candidate scoring and resume-grounded chatbot with structured prompt engineering.
- Designed normalized PostgreSQL schema with GORM for transactional operations.

ClutterFlow — Document AI Platform
Python, FastAPI, React, Gemini LLM, OCR, Supabase, Docker, Cloud Run | GitHub
- Built a full-stack platform for classifying and summarizing unstructured documents using an OCR pipeline integrated with Gemini LLM and structured prompt engineering.
- Deployed backend on Google Cloud Run and frontend on Vercel with Supabase for data persistence.

Distributed Logging & Monitoring System
Apache Kafka, Fluentd, Elasticsearch, Kibana, Docker | GitHub
- Designed a high-throughput log aggregation pipeline using Kafka and the ELK stack for real-time search, query processing, and visualization.
- Containerized the entire distributed architecture using Docker Compose for reproducible deployment.

Multimodal Emotion Recognition System
Python, TensorFlow, PyTorch, librosa, scikit-learn | GitHub
- Engineered multimodal emotion recognition using audio-visual conversational datasets with early and late fusion strategies.

TECHNICAL SKILLS

Languages: Python, C++, C, Java, TypeScript, JavaScript, Go, SQL
ML & AI: PyTorch, TensorFlow, HuggingFace Transformers, scikit-learn, DeBERTa, BERT, LSTM, GRU, LLM, GenAI, RAG, Prompt Engineering, FAISS, BM25
Systems & Data: Apache Kafka, Apache Flink, RabbitMQ, Redis, Elasticsearch, Spark, Hadoop, PostgreSQL, MySQL, Supabase, Prisma ORM
Cloud & DevOps: Docker, Kubernetes, AWS, GCP (Cloud Run), Vercel, Railway, Git, GitHub, CI/CD
Frontend: React, TypeScript, Tailwind CSS, Vite, Streamlit, HTML/CSS

CERTIFICATIONS & AWARDS

- Machine Learning Specialization — DeepLearning.AI
- Anthropic Claude Certification
- Data Structures & Algorithms Certification
- Winner, InGenious 11.0 Hackathon — Disaster Management Robot (2023)
- 3D Animation Workshop Instructor — Taught Blender to ~98 students (Parallax Club)
- Core Member, Parallax Game Development & AR/VR Club
- University Basketball Team (2022–23)